

SE3070 - Case Studies in Software Engineering

Assignment 02 - Proposed Improved Design

UC-01: Issue Hazard Warning

Improved version of the Group_026 design, Case Study 2:
Smart Disaster Early-Warning and Emergency Coordination System for Sri Lanka

Contents: 1. Use case scenario · 2. Use case diagram · 3. Class diagram · 4. Sequence diagram · 5. Wireframe changes · 6. Change justification

1. Corrected Use Case Scenario

Use Case ID	UC-01
Use Case Name	Issue Hazard Warning
Actors	Primary: DMC Officer, Duty Officer (specialises DMC Officer). Secondary: Citizen (recipient), Notification Service.
Description	An authorised officer composes a location-specific hazard warning, previews its reach and broadcasts it to all registered citizens in the target districts or river basins by push notification, SMS and audible alert. Delivery is tracked per recipient and per channel. Active warnings can later be updated or ended with an all-clear.
Trigger	The officer identifies, from the monitoring dashboard, a confirmed ground report (UC-02) or external hazard data, that a hazard warrants public notification.
Preconditions	• The officer is authenticated with warning-issuance privileges. • Districts and river basins are registered, and citizens have a home district on file. • Push, SMS and audible channels are configured in the Notification Service.
Postconditions (success)	• A HazardAlert exists with status BROADCAST (UPDATED or CANCELLED for A2/A3), recording the issuing officer, time and version. • A Notification record exists for every recipient and channel, with its final delivery status. • The alert is available for the post-event report (UC-04).
Postconditions (failure)	The alert remains DRAFT or is discarded. No notification is sent.

Main Flow

1. Officer opens “Issue Hazard Warning” on the DMC dashboard.
2. System opens a new warning with status DRAFT.
3. Officer selects the hazard type (Flood, Landslide, Cyclone, Drought).
4. Officer selects the severity level (Low, Medium, High, Severe).
5. Officer selects one or more districts or river basins as the target scope.
6. System validates the scope against the registered areas and checks for an active warning of the same hazard type in that scope, and finds none.
7. System counts the registered citizens in the scope and generates a preview message from the hazard type and severity.
8. Officer reviews and optionally edits the message (max 160 characters, so it fits in one SMS).
9. Officer selects “Confirm & Broadcast”.
10. System shows a confirmation dialog stating the severity, areas, recipient count and channels.
11. Officer confirms.
12. System sets the alert to BROADCAST and records the issuing officer and time.
13. System sends push, SMS and audible alerts to every recipient through the Notification Service and creates a Notification record for each recipient and channel.
14. System updates each Notification with its delivery result and displays the delivery summary (sent, delivered and failed per channel).
15. Use case ends.

Alternative Flows

A1 - Escalation from a confirmed report (before step 3)

- A1.1 Officer opens a CONFIRMED hazard report (UC-02) and selects “Escalate to Warning”.
- A1.2 System pre-fills the hazard type and the district containing the report location, and links the alert to the report.
- A1.3 Resume at step 4.

A2 - Active warning already exists (step 6)

- A2.1 System shows the active warning and offers “Update existing warning” instead of creating a duplicate.
- A2.2 Officer changes the severity and/or scope.
- A2.3 System sets the status to UPDATED and increments the version. Resume at step 7, sending an update message to the recalculated recipients.

A3 - Issue all-clear

- A3.1 Officer opens an active warning and selects “Issue All-Clear”.
- A3.2 After confirmation, the system sets the status to CANCELLED and sends an all-clear message to the original recipients.
- A3.3 Resume at step 14.

A4 - Officer backs out (step 10)

- A4.1 Officer selects Back in the confirmation dialog and returns to the form, or discards the draft.
- A4.2 Nothing is sent.

Exception Flows

E1 - Invalid target scope (step 6)

- E1.1 No area is selected, or an area does not match a registered district or river basin.
- E1.2 System shows a validation error and highlights the scope field. Resume at step 5.

E2 - No recipients in scope (step 7)

- E2.1 The recipient count is zero.
- E2.2 System blocks the broadcast and tells the officer no registered citizens are in the scope. Resume at step 5.

E3 - Channel delivery failure (step 13)

- E3.1 Some deliveries fail, for example push notifications during a network outage.
- E3.2 System resends each failed alert via the fallback channel (SMS), up to 3 attempts.
- E3.3 Alerts that still fail are marked FAILED, and the summary shows partial delivery with a list of unreached citizens.

2. Corrected Use Case Diagram (UC-01 scope)

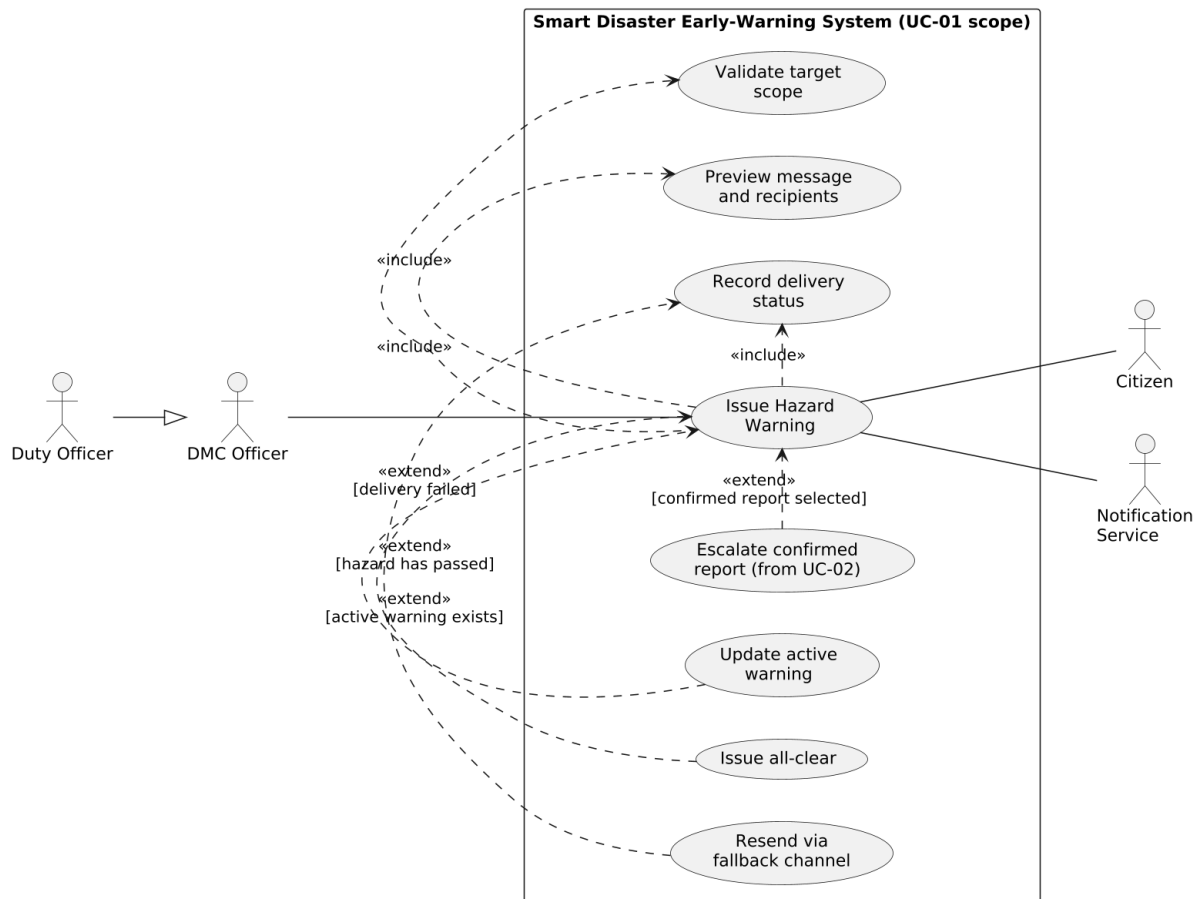


Figure 1: Improved use case diagram for Issue Hazard Warning

Changes from the original

- Removed «include» on Select hazard type, Set severity level and Define target scope. These are steps of the flow, not use cases.
- Removed the per-channel «include»s (Send push / SMS / audible). Channels are a technical detail handled by the Notification Service and the class design.
- Added a generalization from Duty Officer to DMC Officer, so both can issue warnings without duplicate associations.
- Added «include» *Preview message and recipients*, which exists in the hi-fi wireframe but was missing from the diagram.
- Added «extend» *Update active warning* and *Issue all-clear*, so a warning can be escalated and ended.
- Kept «extend» from the escalated confirmed report (UC-02 link) and moved the delivery-failure extension onto *Record delivery status*.

3. Corrected Class Diagram (warning portion)

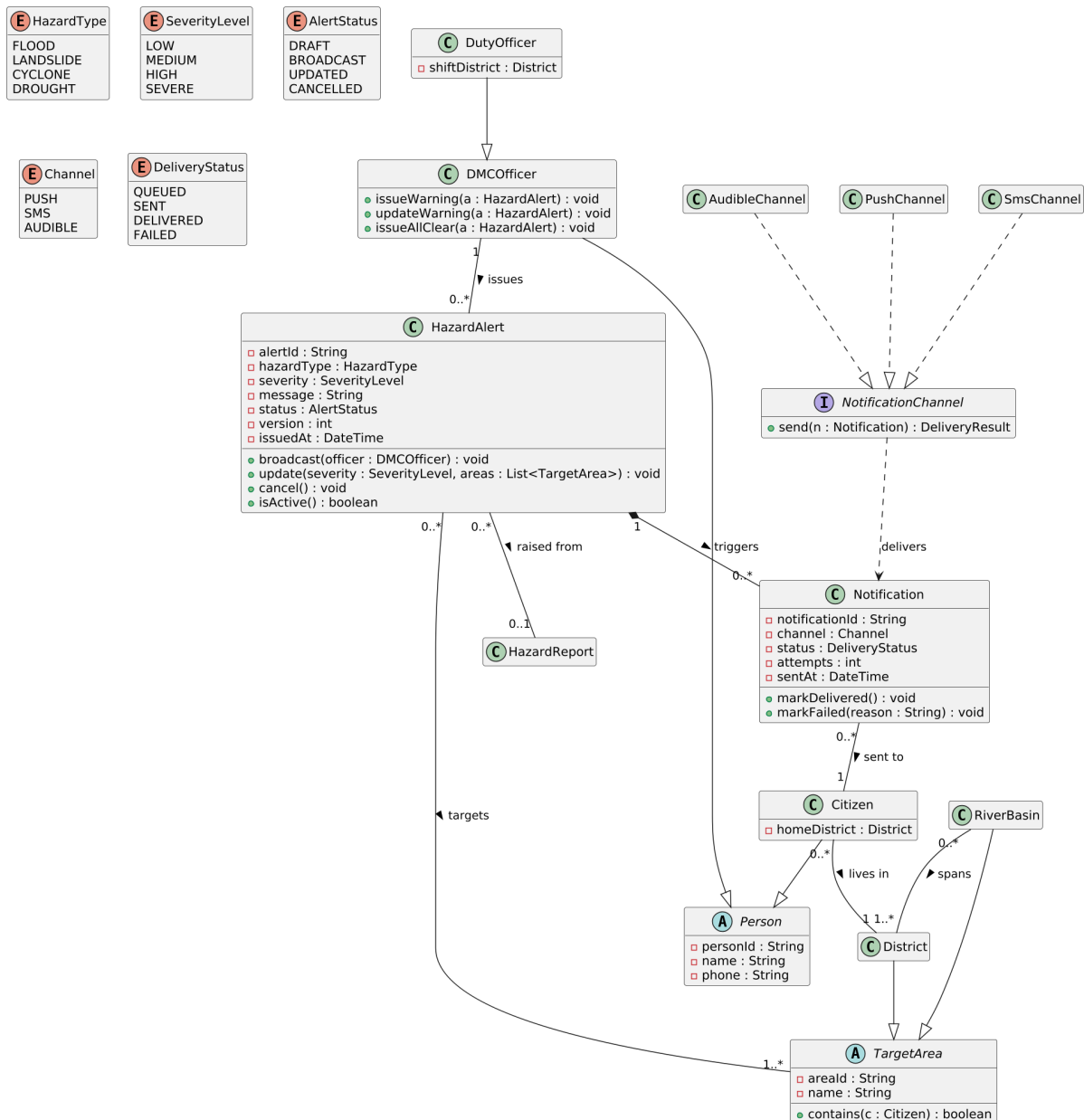


Figure 2: Improved class diagram for hazard warnings and notifications

Changes from the original

- DutyOfficer now specialises DMCOfficer, removing the duplicated issueWarning() method from both classes.
- HazardAlert uses HazardType, SeverityLevel and AlertStatus enums instead of Strings, and adds message and version.
- Replaced District.riverBasin : String with a TargetArea abstraction. A RiverBasin can span several districts, which a String cannot express.
- Notification now records channel, delivery status and attempts per recipient, so delivery can be tracked and retried and UC-04 can count citizens reached.
- Added a NotificationChannel interface with Push, SMS and Audible implementations (Strategy pattern), so a new channel can be added without changing the controller.
- HazardAlert links to its source HazardReport with a plain association, replacing the incorrect «extend» used in the original class diagram.

4. Corrected Sequence Diagram

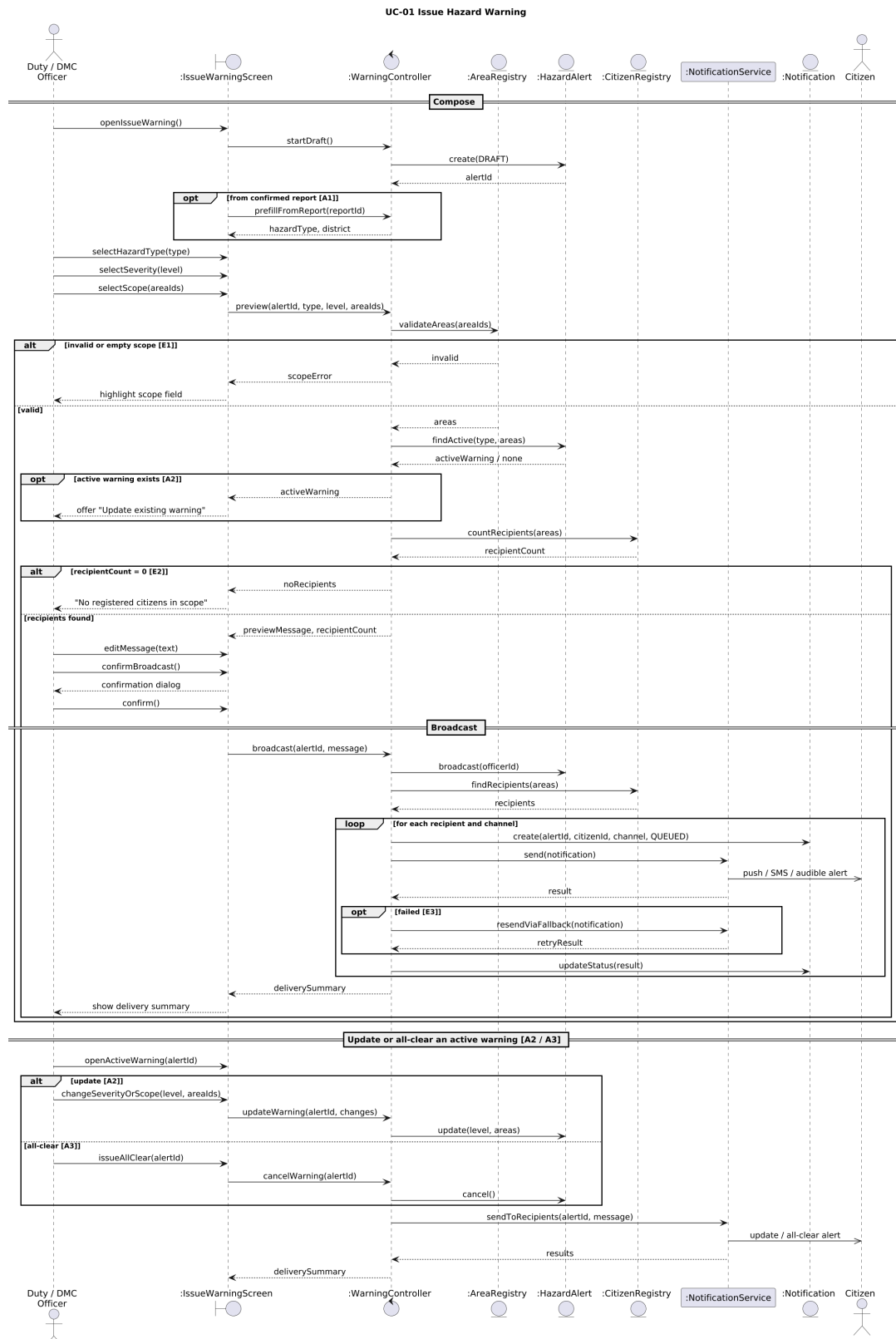


Figure 3: Improved sequence diagram – Issue Hazard Warning

Fixes to the original

- The original sent a single selectHazardType(type, severity, targetArea) message. The diagram now shows the separate steps from the scenario.

- The original opt said “return error, abort” while the scenario says the officer corrects and resumes. E1 now returns an error and the officer corrects the scope.
- Added the missing preview, recipient count and confirmation steps, the A1 pre-fill, and the duplicate-warning check (A2).
- Added a loop over recipients and channels, with a Notification entity recording each delivery.
- The draft is created when composing starts, not after the officer confirms.
- Added a section for updating an active warning and issuing an all-clear.

5. Wireframe Changes

5.1 Issue Hazard Warning (updated)

DMC Command Console - Issue Hazard Warning Duty Officer - Colombo District

(i) Pre-filled from confirmed report GR-2481
(!) An active Flood warning (HIGH) already covers Colombo Update existing

1. Hazard type

Flood

Landslide

Cyclone

Drought

Severity

LOW

MEDIUM

HIGH

SEVERE

2. Target scope

Districts

River basins

Search district...

☒ Colombo

☒ Gampaha

☐ Kalutara

3. Broadcast preview (editable)

Flood Warning: SEVERE. Move to higher ground and follow official guidance.

96/160 characters (one SMS)

Push notification Ready

SMS gateway Ready

Audible alert Ready

48,200 citizens in target scope

Status: DRAFT

Cancel

Confirm & Broadcast

Figure 4: Updated Issue Hazard Warning screen

- Banners for a pre-filled confirmed report (A1) and an existing active warning (A2).
- Separate Districts and River basins tabs with search, instead of one mixed list of four items (Sri Lanka has 25 districts).
- Editable broadcast preview with an SMS character counter.
- The recipient count, already in the original hi-fi, is kept and also added to the lo-fi for consistency.

5.2 Confirmation dialog and delivery summary (new)

Confirm broadcast

You are about to send a **SEVERE Flood** warning to **48,200** citizens in Colombo and Gampaha via Push, SMS and Audible alert.

This cannot be recalled. Use All-Clear to end it.

Back

Broadcast now

Delivery summary - Alert HA-1043 (BROADCAST)

Channel	Sent	Delivered	Failed
Push	48,200	44,910	3,290
SMS	48,200	47,960	240
Audible	31,500	31,380	120

3,050 failed push alerts resent via SMS (fallback)
240 citizens not reached

View list

Update warning

Issue All-Clear

Done

Figure 5: New confirmation dialog and delivery summary screen

- Confirmation dialog before an irreversible, high-impact broadcast (error prevention, Nielsen heuristic 5).
- Delivery summary per channel with fallback results and unreachable citizens (step 14 and E3), which had no screen in the original.
- Update warning and Issue All-Clear actions (A2, A3).

6. Change → Justification

Change	Critique it addresses
Draft created at composition; status enum	The original created a DRAFT only after the officer confirmed, and then immediately escalated it, so the draft had no purpose.
Recipient counting in scenario and sequence	The hi-fi wireframe showed 48,200 citizens, but no step or message produced this number.
Active-warning check (A2) and all-clear (A3)	The original could only create new warnings, allowing duplicates, with no way to escalate or end a warning.
No-recipients exception (E2)	An empty target scope was not handled.
Notification entity with per-channel status	Delivery confirmations were mentioned in step 10 but never modelled, and UC-04 needs them for “citizens reached”.
Steps separated in the sequence diagram	One message bundled three user actions; the invalid-scope opt contradicted the scenario; there was no loop over recipients.
DutyOfficer specialises DMCOfficer	issueWarning() was duplicated in two unrelated classes.
TargetArea / RiverBasin classes	riverBasin as a String on District cannot represent a basin spanning several districts.
NotificationChannel strategy	Channels as a String made adding or retrying a channel require controller changes (Open/Closed principle).
Confirmation dialog and delivery summary screen	An irreversible mass broadcast had no confirmation step, and step 10's summary had no interface design.
Tabs and search for target scope	A single mixed list of districts and basins does not scale to 25 districts and 100+ river basins.

Note for implementation and testing: the alternative flows (A1–A4) and exception flows (E1–E3) map directly to unit tests, for example scope validation, zero recipients, duplicate active warnings, fallback retry limits and status transitions (DRAFT → BROADCAST → UPDATED / CANCELLED).